# Ibrahim Feknous

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## EDUCATION

**NEW JERSEY INSTITUTE OF TECHNOLOGY** GPA: 3.76

*BS/MS ELECTRICAL ENGINEERING* *EXPECTED GRADUATION: December 2027*

- Albert Dorman Honors College Scholar

## EXPERIENCE

**Human Friendly Robotics** April 2026 – August 2026

*Robotics Engineer Intern*

- Built autonomous robots for construction focused on automating repetitive work to improve jobsite productivity and safety
- Designed and routed a custom embedded robotics motherboard integrated with Teensy-based control, RJ45 sensor interfaces, opto-isolated relay channels, and power distribution, employing oscilloscopes, probing techniques, power supplies, and multimeters
- Implemented transistor-driven relay control with flyback protection for noise immunity, switching reliability, and interface safety
- Developed scalable PCB architecture to centralize wiring, simplify debugging, and deployment across projected 80+ robot fleet

**Pure Power Engineering** June 2026 – August 2026

*Electrical Engineer I Intern*

- Designed and supported commercial solar PV and energy storage systems ranging from 100 kW to 20 MW
- Developed AutoCAD engineering drawings, construction documents, and layouts for PV projects under NFPA 70 and IBC code
- Prepared utility interconnection and permitting packages by collecting technical data and coordinating project requirements
- Performed PVsyst production modeling, site analysis, and field visits to support design validation and implementation

**NJIT Ubiquitous Networking and Intelligent Computing System Lab** May 2025 – August 2026

*Undergraduate Researcher*

- Engineered an edge-to-cloud computing pipeline using a TRAXXAS vehicle platform to reduce latency and computation cost
- Validated autonomous system performance through CARLA simulations, demonstrating a 70% improvement in response time
- Collaborated with PhD researchers, cross-disciplinary faculty, and Siemens research engineers through a 10-week research program

## PROJECTS

**NJIT Solar Car Team** | *President, Re-Founder* *February 2025 - Present*

- Re-founded and led NJIT's Solar Car Team, managing 20+ board members and 60+ development team members across all sub-teams
- Drive full-vehicle development across battery management, fire suppression, power distribution for low and high voltage, central control system, embedded display, autonomous system, composite body, steering and suspension, wheels and brakes, and chassis
- Developed sponsorship and university engagement strategy contributing to $120,000+ in sponsorships for the NJIT's program
- Built project structures enabling capstone, research credit, and cross-functional engineering development for students

**STM32-Based Brushless Motor Controller with Regeneration** | *Senior Capstone Project* *August 2026 - Present*

- Developing a regenerative STM32-based traction inverter in Altium to drive a 3 kW BLDC motor from a 72 V battery pack using FOC and SVPWM
- Implementing PWM generation, current sensing, protection logic, fault detection, and CAN/Ethernet communication for real-time motor control optimized for EV-related student team use
- Designing a product-style interface for current limits, operating parameters, telemetry, fault-code reporting, and automatic shutdown

**Open-Loop Synchronous Buck Converter** | *Electronic Circuit Design Project* *April 2026 – May 2026*

- Developed a hardware-controlled synchronous DC-DC buck converter using PWM duty-cycle control, bootstrap high-side gate driving, and dual MOSFET switching for efficient step-down voltage
- Designed and routed the printed circuit board in KiCad, validating output ripple, load behavior, and switching performance through LTspice simulation across multiple current loads

**NJIT IEEE Geoscience and Remote Sensing Society** | *President* *January 2026 – Present*

- Leading NJIT's high-altitude balloon project to measure cosmic ray activity and atmospheric conditions at 110,000+ feet
- Designing an STM32-based sensor PCB with LoRa telemetry, GPS, and SPI/I²C for payload data collection and wireless communication

## EXTRACURRICULARS

**NJIT IEEE Student Branch** January 2025 – Present

*Vice-President, Executive Chair, Conference Chair*

- Serve as Vice President of NJIT IEEE Student Branch while leading large-scale technical programming, sponsorship outreach, and student engineering initiatives
- Executive Chair for MakeNJIT Hardware Hackathon, directing logistics, outreach, marketing, finance, and sponsor relations for a 250+ participant hardware-focused engineering event bringing in over $80,000+ in sponsorships
- Conference Chair for the 2027 IEEE Region 1 & 2 Student Activities Summit with a $50,000+ sponsorship target and 250+ projected student participation from 10+ states, hosting various workshops and competitions in multiple disciplines of electrical engineering

## SKILLS

**TECHNICAL:** PCB Design, Power Electronics, Embedded Systems, Solar PV, IoT, Robotics, Computer Vision, CNC Machining, Soldiering, KiCad, AutoCAD, PVsyst, NI Multisim, Fusion 360, SolidWorks, OnShape, Revit, MATLAB, Simulink, Python, C/C++, Java, RISC-V, Ubuntu, Arduino IDE, CARLA, LTspice, Altium, Altair, Siemens PSSE, BlueBeam, Microsoft Suite